SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA1613 COURSE CODE: Data Warehousing and Data Mining

---

COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3,BL4,BL5)
Assessment Tool Weightage: 30%

QUESTIONS: 3 **Total Marks:30**

Annexure C

Case study

Criteria	Understanding of Case ***(2.5)***	Application of Concepts ***(2.5)***	Analysis ***(2)***	Solution Design ***(2)***	Clarity ***(1)***	***Total(10)***
Weightage	*25%*	25%	20%	20%	10%	*100%*
Q1						
Q2						
Q3						

Code of Conduct:

*I am **J.Ramesh (192311230)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Ramesh

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, wellstructured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

ANNEXURE A

Case Study:

1. A shepherd boy gets bored tending the town's flock. To have some fun he cries out, "Wolf!" even though no wolf is in sight. The villagers run to protect the flock, but then get really mad when they realize the boy was playing a joke on them.

One night, the shepherd boy sees a real wolf approaching the flock and calls out, "Wolf!" The villagers refuse to be fooled again and stay in their houses. The hungry wolf turns the flock into lamb chops. The town goes hungry. Panic ensues. Identify TP,FP,FN,TN

2.Assume there are 100 images, 30 of them depict a cat, the rest do not. A machine learning model predicts the occurrence of a cat in 25 of 30 cat images. It also predicts absence of a cat in 50 of the 70 no cat images. Identify TP,FP,FN,TN and Calculate the Precision and Recall and F1 Score .

3.An email service provider wants to classify emails as **Spam or Not Spam** using features:

- Contains “Offer” (Yes/No)
- Contains “Free” (Yes/No)
- Email Length (Short/Long)

Sample Dataset

Offer	**Free**	**Length**	**Class**
Yes	Yes	Short	Spam
Yes	No	Long	Not Spam
No	Yes	Short	Spam
No	No	Long	Not Spam
Yes	Yes	Long	Spam
No	No	Short	Not Spam

Questions

1. Calculate prior probabilities:
    - P(Spam), P(Not Spam)
2. Compute conditional probabilities for each feature.
3. Using **Naïve Bayes**, classify:
    - (Offer=Yes, Free=No, Length=Short)
4. Compare results using **WEKA (Naïve Bayes classifier)**.
5. Discuss the **independence assumption** in Naïve Bayes

Answers

1.Confusion Matrix for the Shepherd Boy and Wolf

Problem We can consider:

- **Positive (+)** = A wolf is actually present / a "Wolf!" warning is given.
- **Negative (−)** = No wolf is present / no "Wolf!" warning is given.
- **Actual condition** = Whether a wolf is really present.
- **Prediction** = Whether the boy gives a "Wolf!" warning.

	Wolf Present (Actual +)	**Wolf Absent (Actual −)**
"Wolf!" Warning (+)	**TP**	**FP**
No "Wolf!" Warning (−)	**FN**	**TN**

1. True Positive (TP):

A **real wolf is approaching the flock**, and the shepherd boy correctly cries **"Wolf!"**.

- Actual condition: Wolf is present.
- Prediction/warning: Wolf is present.
- The warning is correct.

Therefore:

TP = Real wolf + Warning given

2. False Positive (FP):

There is **no wolf**, but the shepherd boy cries **"Wolf!"** as a joke.

- Actual condition: No wolf.
- Prediction/warning: Wolf is present.
- The warning is incorrect. Therefore:

FP = No wolf + False warning

This happens when the villagers run to protect the flock even though there is no danger.

3. False Negative (FN)

A **real wolf is present**, but the warning is **not recognized or acted upon**.

In the story, the boy actually calls "Wolf!" when the real wolf appears, but the villagers ignore the warning because of his previous false alarms.

For the classification interpretation, this represents a **missed detection**:

- Actual condition: Wolf is present.
- Effective prediction/action: No warning response.
- The real danger is missed.

Therefore:

FN = Real wolf + Danger missed

4. True Negative (TN):

There is **no wolf**, and the boy **does not give a false "Wolf!" warning**.

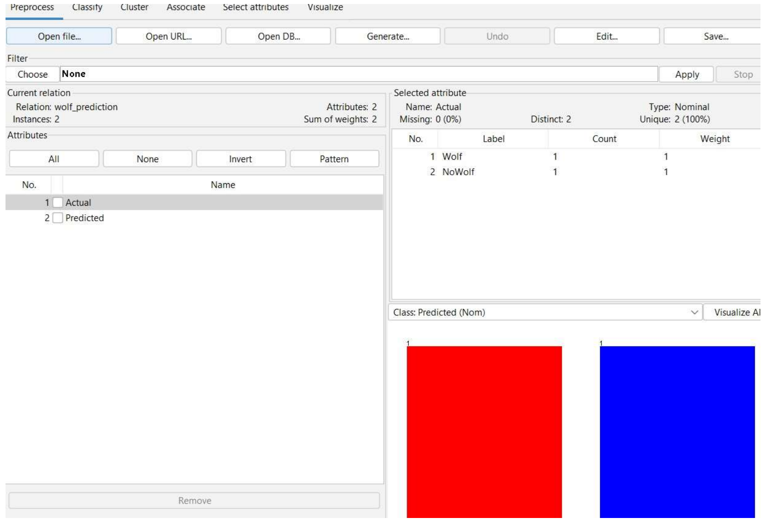
- Actual condition: No wolf.
- Prediction: No wolf.
- The decision is correct.

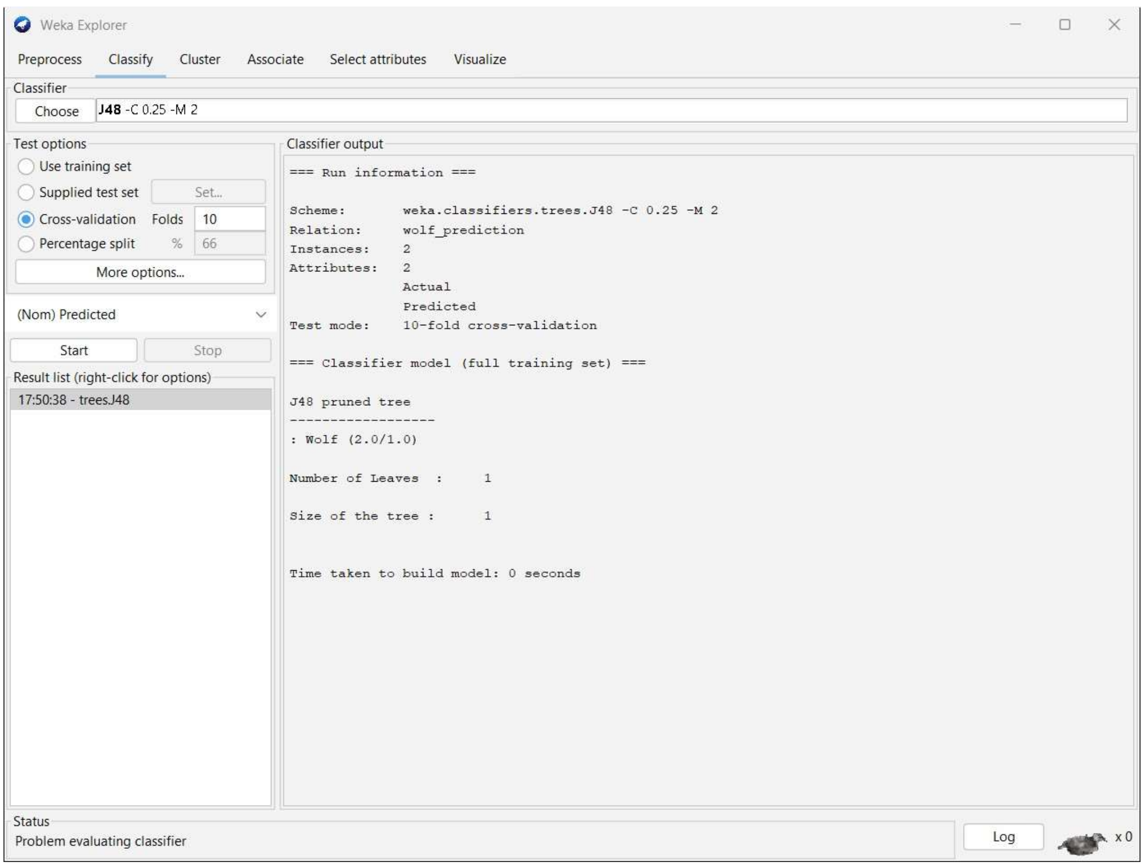
Therefore:

TN = No wolf + No warning

Final Answer:

Term	**Meaning in this problem**
TP	Wolf is present and the warning correctly identifies it
FP	No wolf is present, but the boy falsely cries "Wolf!"
FN	Wolf is present, but the danger is missed/not effectively detected
TN	No wolf is present and no false warning is given


Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Open file...
Open URL...
Open DB...
Generate...
Undo
Edit...
Save...
Filter
Choose
None
Apply
Stop
Current relation
Relation: wolf_prediction
Instances: 2
Attributes: 2
Sum of weights: 2
Selected attribute
Name: Actual
Missing: 0 (0%)
Distinct: 2
Type: Nominal
Unique: 2 (100%)
Attributes
All
None
Invert
Pattern
No.
Name
1 Actual
2 Predicted
No. Label Count Weight
1 Wolf 1 1
2 NoWolf 1 1
Class: Predicted (Nom)
Visualize All
Remove


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Classifier
Choose
J48 -C 0.25 -M 2
Test options
Use training set
Supplied test set
Set...
Cross-validation Folds 10
Percentage split % 66
More options...
(Nom) Predicted
Start
Stop
Result list (right-click for options)
17:50:38 - trees.J48
Classifier output
=== Run information ===
Scheme: weka.classifiers.trees.J48 -C 0.25 -M 2
Relation: wolf_prediction
Instances: 2
Attributes: 2
Actual
Predicted
Test mode: 10-fold cross-validation
=== Classifier model (full training set) ===
J48 pruned tree

: Wolf (2.0/1.0)
Number of Leaves : 1
Size of the tree : 1
Time taken to build model: 0 seconds
Status
Problem evaluating classifier
Log
x 0

2. Confusion Matrix, Precision, Recall and F1 Score

Given:

- **Total images = 100**
- **Images with cats = 30**
- **Images without cats = 70**
- Model correctly predicts a cat in **25 of the 30 cat images**
- Model correctly predicts no cat in **50 of the 70 non-cat images**

Step 1: Identify TP, FN, TN, and FP:

True Positive (TP):

The image actually contains a cat, and the model correctly predicts a cat.

TP = 25

False Negative (FN):

The image actually contains a cat, but the model predicts no cat.

There are 30 cat images, and 25 were correctly identified:

FN = 30 − 25 = 5

True Negative (TN):

The image does not contain a cat, and the model correctly predicts no cat.

TN = 50

False Positive (FP):

The image does not contain a cat, but the model predicts that there is a cat.

There are 70 non-cat images, and 50 were correctly identified:

FP = 70 − 50 = 20

Confusion Matrix:

	Actual Cat	**Actual No Cat**
Predicted Cat	**TP = 25**	**FP = 20**
Predicted No Cat	**FN = 5**	**TN = 50**

So:

- **TP = 25**
- **FP = 20**

- **FN = 5**
- **TN = 50**

Step 2: Calculate Precision:

Precision tells us **how many of the images predicted as cats were actually cats**.

Formula:

Formula:

$$Precision = \frac{TP}{TP + FP}$$

Substitute the values:

$$Precision = \frac{25}{25 + 20}$$

$$Precision = \frac{25}{45}$$

$$\boxed{Precision = 0.556}$$

or

$$\boxed{Precision = 55.56\%}$$

Step 3: Calculate Recall:

Recall tells us **how many of the actual cat images were correctly detected.**

Formula:

$$Recall = \frac{TP}{TP + FN}$$

$$Recall = \frac{25}{25 + 5}$$

$$Recall = \frac{25}{30}$$

$$\boxed{Recall = 0.8333}$$

or

$$\boxed{Recall = 83.33\%}$$

Step 4: Calculate F1 Score:

F1 Score is the harmonic mean of Precision and Recall.

Formula:

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

Substitute:

$$F1 = \frac{2 \times 0.556 \times 0.8333}{0.556 + 0.8333}$$

$$F1 \approx 0.667$$

Therefore:

$$\boxed{F1 Score \approx 0.667}$$

or

$$\boxed{F1 Score \approx 66.67\%}$$

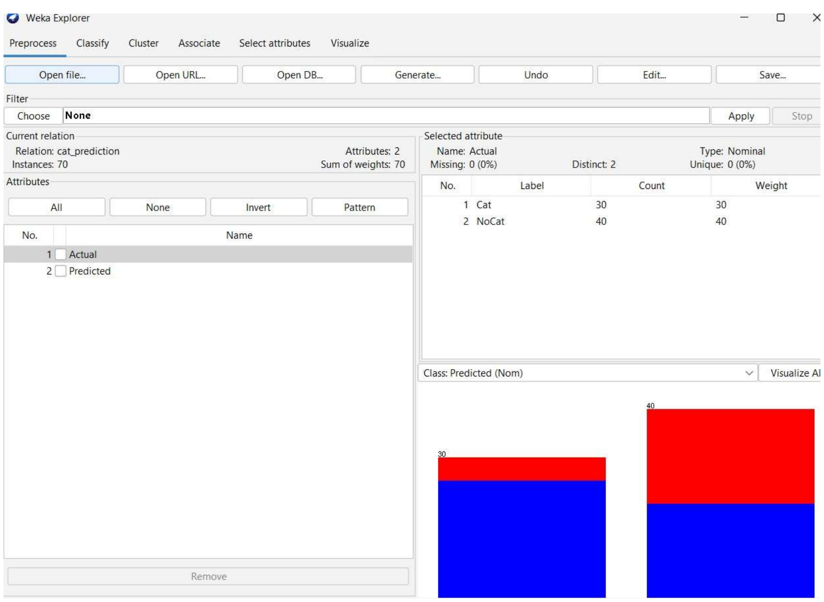
Final Answer:

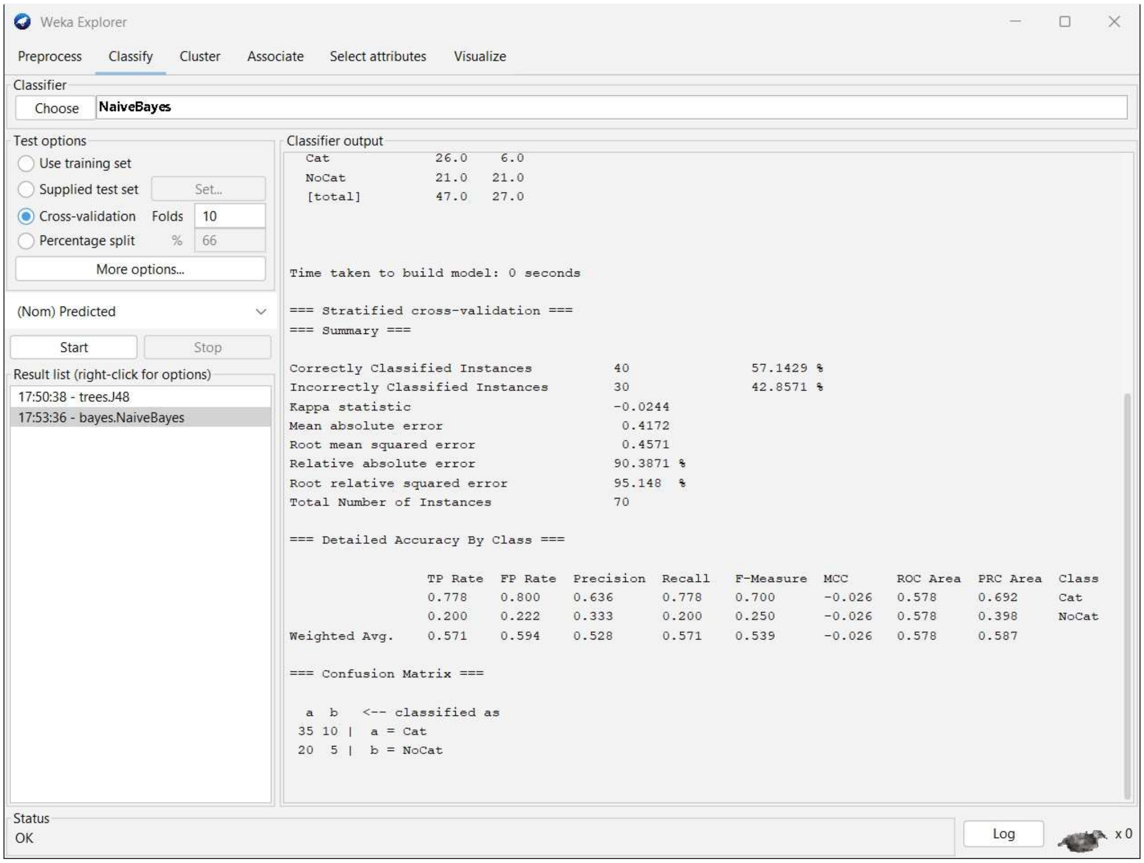
Measure Value

TP	**25**
FP	**20**
FN	**5**
TN	**50**
Precision	**55.56%**
Recall	**83.33%**
F1 Score	**66.67%**

Conclusion:

The model has a **high recall (83.33%)**, meaning it detects most of the actual cat images, but its **precision is lower (55.56%)**, meaning many of its cat predictions are incorrect.





3. Naïve Bayes Classification – Spam Email

Given Dataset

Offer	**Free**	**Length**	**Class**
Yes	Yes	Short	Spam
Yes	No	Long	Not Spam
No	Yes	Short	Spam
No	No	Long	Not Spam
Yes	Yes	Long	Spam
No	No	Short	Not Spam

Total emails = **6**

- Spam = **3**
- Not Spam = **3**

1. **Calculate Prior Probabilities:**

The formula is:

$$P(Class) = \frac{\text{Number of emails in Class}}{\text{Total emails}}$$

Probability of Spam

$$P(Spam) = \frac{3}{6} = 0.5$$

Probability of Not Spam

$$P(NotSpam) = \frac{3}{6} = 0.5$$

Therefore:

$$\boxed{P(Spam) = 0.5}$$

$$\boxed{P(NotSpam) = 0.5}$$

2. **Calculate Conditional Probabilities**

We need to calculate the probability of each feature value given the class.

A. Feature: Offer

For Spam

There are 3 Spam emails:

- Offer = Yes → 2
- Offer = No → 1 Therefore:

$$P(Offer = Yes|Spam) = \frac{2}{3} = 0.667$$

$$P(Offer = No|Spam) = \frac{1}{3} = 0.333$$

For Not Spam

There are 3 Not Spam emails:

- Offer = Yes → 1
- Offer = No → 2 Therefore:

$$P(Offer = Yes|NotSpam) = \frac{1}{3} = 0.333$$

$$P(Offer = No|NotSpam) = \frac{2}{3} = 0.667$$

B. Feature: Free

For Spam:

Among the 3 Spam emails:

- Free = Yes → 2
- Free = No → 1 Therefore:

$$P(Free = Yes|Spam) = \frac{2}{3} = 0.667$$

$$P(Free = No|Spam) = \frac{1}{3} = 0.333$$

For Not Spam:

Among the 3 Not Spam emails:

- Free = Yes → 0
- Free = No → 3 Therefore:

$$P(Free = Yes|NotSpam) = \frac{0}{3} = 0$$

$$P(Free = No|NotSpam) = \frac{3}{3} = 1$$

C. Feature: Length

For Spam

Among the 3 Spam emails:

- Short → 2
- Long → 1 Therefore:

$$P(Length = Short|Spam) = \frac{2}{3} = 0.667$$

$$P(Length = Long|Spam) = \frac{1}{3} = 0.333$$

For Not Spam

Among the 3 Not Spam emails:

- Short → 1
- Long → 2 Therefore:

$$P(Length = Short|NotSpam) = \frac{1}{3} = 0.333$$

$$P(Length = Long|NotSpam) = \frac{2}{3} = 0.667$$

Conditional Probability Table:

Feature	**Spam**	**Not Spam**
P(Offer=Yes)	2/3	1/3
P(Offer=No)	1/3	2/3
P(Free=Yes)	2/3	0
P(Free=No)	1/3	1
P(Length=Short)	2/3	1/3
P(Length=Long)	1/3	2/3

3. Classify (Offer=Yes, Free=No, Length=Short)

We need to calculate the probability that the email is **Spam** and **Not Spam**.

The Naïve Bayes formula is:

$$P(Class|Features) \propto P(Class) \times P(Offer|Class) \times P(Free|Class) \times P(Length|Class)$$

Probability of Spam

Given:

- $P(Spam) = 1/2$
- $P(Offer = Yes|Spam) = 2/3$
- $P(Free = No|Spam) = 1/3$
- $P(Length = Short|Spam) = 2/3$

Therefore:

$$P(Spam|X) \propto \frac{1}{2} \times \frac{2}{3} \times \frac{1}{3} \times \frac{2}{3}$$
$$= \frac{4}{54}$$
$$= \boxed{0.0741}$$

Probability of Not Spam:

Given:

- $P(NotSpam) = 1/2$
- $P(Offer = Yes|NotSpam) = 1/3$
- $P(Free = No|NotSpam) = 1$
- $P(Length = Short|NotSpam) = 1/3$

Therefore:

$$P(NotSpam|X) \propto \frac{1}{2} \times \frac{1}{3} \times 1 \times \frac{1}{3}$$
$$= \frac{1}{18}$$
$$= \boxed{0.0556}$$

Since:

$$0.0741 > 0.0556$$

the model predicts:

$$\boxed{\text{Spam}}$$

Normalized probabilities:

Total:

$$0.0741 + 0.0556 = 0.1297$$

Spam:

$$P(Spam|X) = \frac{0.0741}{0.1297} \approx 0.5714$$

Not Spam:

$$P(NotSpam|X) = \frac{0.0556}{0.1297} \approx 0.4286$$

Therefore:

$$\boxed{P(Spam|X) = 57.14\%}$$

$$\boxed{P(NotSpam|X) = 42.86\%}$$

Final classification:

$$\boxed{\text{The email is classified as SPAM}}$$

4. Compare Results Using WEKA

In **WEKA**, the same dataset can be entered into an ARFF file and the **NaiveBayes** classifier can be selected.

Expected WEKA result

For:

Offer = Yes, Free = No, Length = Short the Naïve

Bayes classifier should classify the email as:

Spam

Our manual calculation also gives:

Spam = 57.14%

Not Spam = 42.86%

Therefore, the manual result and WEKA classification should agree, assuming the same dataset and settings are used.

Note: Because this is a very small dataset, WEKA's exact displayed probabilities can differ slightly depending on its probability estimation/smoothing settings. The predicted class should remain **Spam** for this example.

5. Independence Assumption in Naïve Bayes

Naïve Bayes assumes that all features are **conditionally independent given the class**.

For this problem, it assumes that:

- Offer
- Free
- Length

are independent of each other once we know whether the email is **Spam** or **Not Spam**.

For example:

is approximated as:

This assumption makes Naïve Bayes **simple and fast**.

Limitation

In real emails, these features may not actually be independent.

For example:

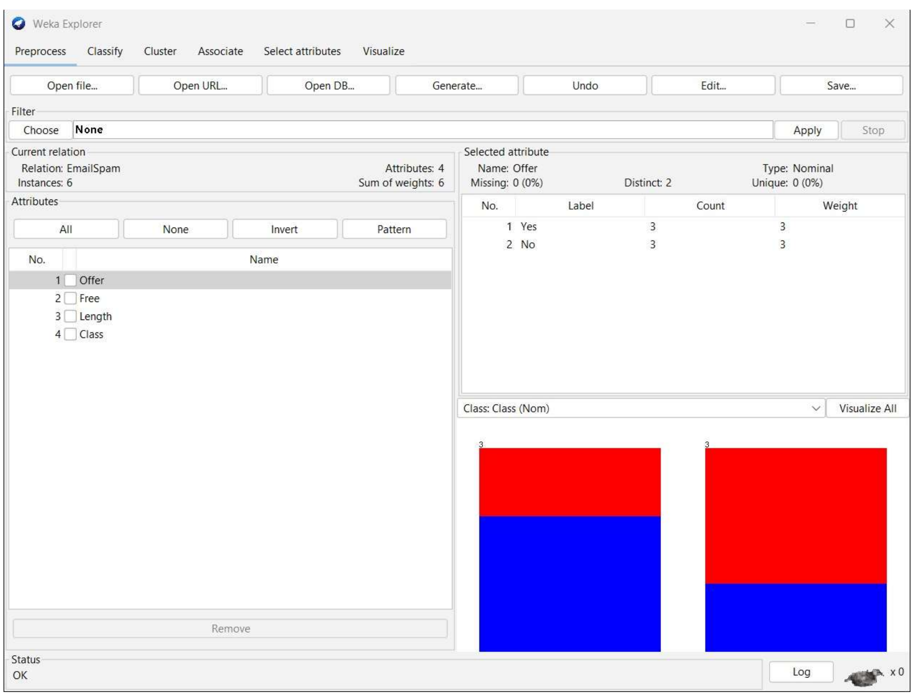
- Emails containing **"Free"** may frequently also contain **"Offer"**.
- Email length may be related to the words used in the email.
- Spam emails may have combinations of features rather than independent features.

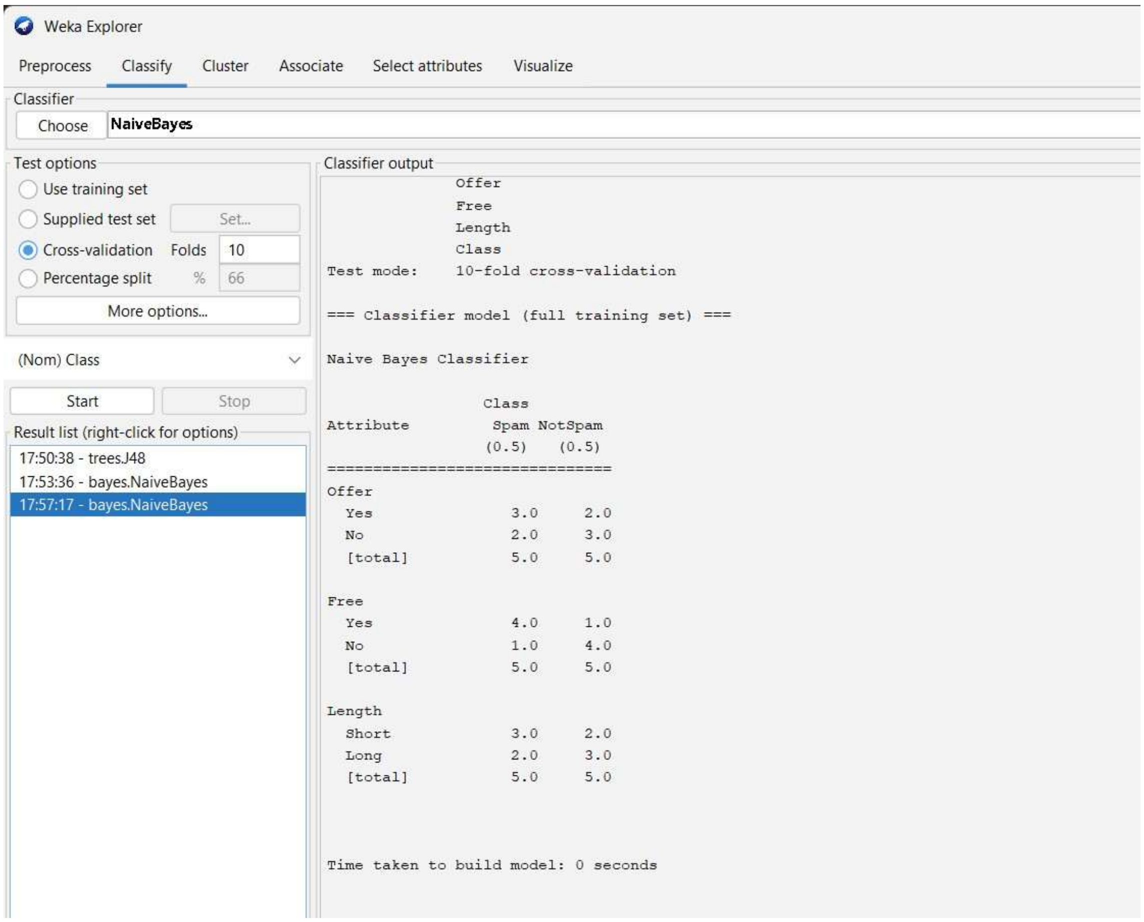
Therefore, the independence assumption may not always be completely realistic. However, **Naïve Bayes can still perform well even when the independence assumption is not perfectly true**.

Final Answers:

Question	**Answer**

P(Spam)	**0.5 (50%)**
P(Not Spam)	**0.5 (50%)**
TP/Conditional probabilities	Calculated above
Spam score for given email	**0.0741**
Not Spam score	**0.0556**
Normalized Spam probability	**57.14%**
Normalized Not Spam probability	**42.86%**
Final classification	**SPAM**
WEKA result	**SPAM**
Independence assumption	Features are assumed conditionally independent given the class




Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Classifier
Choose
NaiveBayes
Test options
Use training set
Supplied test set
Set...
Cross-validation
Folds
10
Percentage split
%
66
More options...
(Nom) Class
Start
Stop
Result list (right-click for options)
17:50:38 - trees.J48
17:53:36 - bayes.NaiveBayes
17:57:17 - bayes.NaiveBayes
Classifier output
Offer
Free
Length
Class
Test mode: 10-fold cross-validation
=== Classifier model (full training set) ===
Naive Bayes Classifier
Class
Attribute Spam NotSpam
(0.5) (0.5)
===============================
Offer
Yes 3.0 2.0
No 2.0 3.0
[total] 5.0 5.0
Free
Yes 4.0 1.0
No 1.0 4.0
[total] 5.0 5.0
Length
Short 3.0 2.0
Long 2.0 3.0
[total] 5.0 5.0
Time taken to build model: 0 seconds